

Practice Quiz: Systems

Part A: Multiple Choice

1. What does the `true_A` matrix in a `DiscreteEnvironment` represent? A) The agent's beliefs about observations B) The true probability of observing o_t given hidden state s_t C) The transition dynamics between states D) The agent's prior over initial states
2. If `true_A` has shape `(5, 3)`, the environment has: A) 5 states and 3 observations B) 3 states and 5 observations C) 5 observations and 3 states D) 15 state-observation pairs
3. When `env.step(action)` is called, what happens first? A) An observation is generated from the current state B) The hidden state transitions according to `true_B` C) The agent updates its beliefs D) Free energy is computed
4. A fully observable environment corresponds to: A) `true_A` being all zeros B) `true_A` being the identity matrix C) `true_B` being the identity matrix D) `true_A` having uniform columns
5. If `true_B` is passed as a 2-D matrix (shape `(N, N)`), the environment assumes: A) There are N possible actions B) There is exactly 1 action C) The transitions are stochastic D) The B matrix is invalid
6. After calling `env.reset(initial_state=0)` followed by 10 calls to `env.step()`, how many entries does `env.history["states"]` contain? A) 10 B) 11 C) 9 D) 12
7. The key difference between `DiscreteEnvironment` and `GenerativeModel` is: A) `DiscreteEnvironment` uses continuous states B) `GenerativeModel` represents ground truth; `DiscreteEnvironment` is an approximation C) `DiscreteEnvironment` represents ground truth; `GenerativeModel` is the agent's hypothesis D) There is no difference; they use the same matrices

Part B: Short Answer

1. Write Python code to create a `DiscreteEnvironment` for a 3-state system where observations are deterministic (identity A-matrix) and action 0 keeps the state while action 1 cycles through states. Show the `true_A` and `true_B` matrices.
2. Explain why columns (not rows) of the `true_A` matrix must sum to 1.0. What probability is each column encoding, and what would go wrong if the normalization were over rows instead?
3. You observe the sequence $[0, 0, 1, 0, 1]$ from a 2-state environment. Using Bayes' rule and the `true_A` matrix $\begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$, compute $P(s_0 = 0 \mid o_0 = 0)$ assuming a uniform prior $P(s_0) = [0.5, 0.5]$.